

Predictors of END in Multivariable analysis

	Clinical (Multivariable Analysis)	Radiological (Multivariable Analysis)
Boulenoir et al. 2021	None	- Suprabulbar iCAo pre-IVT: OR 4.0 (95% CI 1.3–12.2, P=0.015), the only one retained in the model after stepwise variable selection - Post-IVT: none
Boulenoir et al. 2022	END24h - IVT: aHR 2.01 (95% CI 1.07–3.92, P=0.03), adjusted for baseline NIHSS and onset-to-imaging time. END24h occurred earlier in IVT group (median imaging-to-END24h: 2.4h IQR 1.8–7.3 vs 8.3h IQR 4.0–15.3, P=0.04) - Non-atheromatous etiology: END24h less frequent in atheromatous cause (14%) vs cardioembolic (50%), dissection (31%), other/undetermined (35%), P<0.01 NB → the study assessed time-to-END (when will END occur), not END occurrence (if END will occur)	None
Broccolini et al. 2024	AFib: aOR 3.547 (95% CI 1.014–12.406, P=0.048), adjusted for age, antiplatelet, anticoagulant, dominant M2, Menon score, thrombolysis	None
Gwak et al. 2021	IVT - aOR 7.68 (95% CI 2.52–23.43, P≤0.001) in Model 1, adjusted for Tmax >6 seconds ≥53.73 mL, onset-to-door time, IV tPA, and initial DWI volume - aOR 5.97 (95% CI 2.05–17.40, P≤0.001) in Model 2, adjusted for Tmax >6 seconds – DWI ≥32.77 mL, onset-to-door time, IV tPA, and initial DWI volume - aOR 14.94 (95% CI 4.18–53.33, P≤0.001) in Model 3, adjusted for Tmax 4–6 seconds ≥55.20 mL, onset-to-door time, IV tPA, and initial DWI volume	- Tmax >6s ≥53.73 mL (critical hypoperfusion): aOR 12.78 (95% CI 3.36–48.65, P<0.001); Model 1, adjusted for Tmax >6 seconds ≥53.73 mL, onset-to-door time, IV tPA, and initial DWI volume - Mismatch volume ≥32.77 mL (Tmax >6s - DWI): aOR 5.73 (95% CI 2.04–16.08, P=0.001); Model 2, adjusted for Tmax >6 seconds – DWI ≥32.77 mL, onset-to-door time, IV tPA, and initial DWI volume - Tmax 4–6s ≥55.20 mL (mild hypoperfusion): aOR 9.13 (95% CI 2.76–30.17, P<0.001); Model 3, adjusted for Tmax 4–6 seconds ≥55.20 mL, onset-to-door time, IV tPA, and initial DWI volume
Gwak et al. 2022	IVT independently associated with END in both Model 1 and Model 2 (exact aOR values not reported in this paper; see Gwak et al. 2021 for statistics)	Poor perfusion profile - Defined as Wilcoxon-Mann-Whitney generalised OR >1.052 (Model 1): aOR 13.42 (95% CI: 4.38–41.15, P<0.001) - Defined as Wilcoxon-Mann-Whitney generalised OR >1.000 (Model 2): aOR 7.31 (95% CI 2.48-21.49, P<0.001) Perfusion profile = severity-weighted shift of Tmax strata distribution (Tmax 2–4s, 4–6s, 6–8s, 8–10s, >10s proportions vs Tmax>2s total), calculated via Wilcoxon-Mann-Whitney generalised OR comparing individual to cohort. Cutoffs based on Youden's index.
He et al. 2021	Higher baseline glucose: - aOR 1.02 (95% CI 1.01–1.03, P=0.009) in Model 1 (adjusted for SVS as continuous value, glucose, hypertension, smoking, cardiac embolism, undetermined stroke) - aOR 1.01 (95% CI 1.00–1.03, P=0.021) in Model 2 (adjusted for SVS ≥9.45 mm dichotomized, glucose, hypertension, smoking, cardiac embolism, undetermined stroke)	- Longer SVS: aOR 2.03 (95% CI 1.45–2.84, p<0.001); Model 1, adjusted for SVS value (continuous), baseline glucose, hypertension, current smoking, cardiac embolism, undetermined stroke) - SVS ≥9.45 mm: aOR 5.41 (95% CI 1.00–29.27, p=0.001); Model 2, adjusted for SVS ≥ 9.45 mm (dichotomized), baseline glucose, hypertension, current smoking, cardiac embolism, undetermined stroke)
Heitkamp et al. 2025	Higher pre-stroke mRS: aOR 1.42 (95% CI 1.13–1.78, P=0.003), Adjusted for NIHSS at admission, pre stroke mRS, AFib, ICA occlusion, time from admission to groin puncture, tPA, general anesthesia, number of passes, successful recanalization TICl 2b-3, adverse events during EVT, ICH after 24h	- Longer admission-to-groin-puncture time: aOR 1.04 (95% CI 1.02–1.07, P=0.001) per minute - General anesthesia: aOR 1.68 (95% CI 1.08–2.63, P=0.022) - Higher number of passes: aOR 1.15 (95% CI 1.03–1.29, P=0.012) - Adverse events during EVT: aOR 1.89 (95% CI 1.19–3.01, P=0.007) - ICH within 24h: aOR 3.40 (95% CI 1.90–6.07, P<0.001) Adjusted for NIHSS, pre-stroke mRS, AFib, ICA occlusion, admission-to-groin time, tPA, general anesthesia, no. of passes, TICl 2b-3, adverse events during EVT, ICH within 24h - Of note, successful recanalization TICl 2b-3 was protective: aOR 0.29 (95% CI 0.17–0.50, P<0.001)
Hu et al. 2025	Delayed antiplatelet therapy (>24h after IVT): aOR 1.46 (95% CI 1.17–1.84, P<0.001), adjusted for Tmax >4s volume, delayed antiplatelet therapy and occlusion site	Isolated ICA occlusion: aOR 2.49 (95% CI 1.97–4.41, P<0.001), adjusted for Tmax >4s volume, delayed antiplatelet therapy and occlusion site
Itabashi et al. 2023	No multivariable analysis done	No multivariable analysis done
Kim et al. 2022	- History of Hypertension: aOR 1.49 (95% CI 1.03–2.14) - Higher baseline NIHSS: aOR 1.12 per 1-point increase (95% CI 1.02–1.23) Adjusted for age, male sex, occluded artery, NIHSS score, LKW to arrival, Hypertension, Diabetes, Advanced WMH, Baseline ASPECTS, collateral grade at baseline	- Extracranial ICA occlusion (without tandem): aOR 1.72 (95% CI 1.09–2.73). - Tandem occlusion: aOR 2.79 (95% CI 1.57–4.95) Adjusted for age, male sex, occluded artery, NIHSS score, LKW to arrival, Hypertension, Diabetes, Advanced WMH, Baseline ASPECTS, collateral grade at baseline
Lee et al. 2020	No multivariable analysis done	No multivariable analysis done
Mazya et al. 2018	No multivariable analysis done	No multivariable analysis done
Min et al. 2020	None	- Insular lesions, compared with no insular lesions, were independently associated with END with infarct growth (OR 2.75, 95% CI 1.18–6.38, p = 0.02), and END-2 with infarct growth (OR 2.52, 95% CI 1.08–5.87, p = 0.03); included confounders: age, male sex, baseline NIHSS Notes - insular lesions associated with END with infarct growth, and not the other way around! - END was classified by presumed mechanism: END with infarct growth, END with new lesions, and END with swelling.
Saleem et al. 2020	None	None
Seners et al. 2021	None	- More proximal occlusion (vs M2 reference): - distal M1 aOR 2.6 (1.4–5.0), proximal M1 aOR 4.2 (1.8–9.4), tandem aOR 5.1 (2.6–10.3), ICA-T/L aOR 21.0 (8.2–54.2) in Model 1, adjusted for sex, NIHSS, SBP, on-site EVT facility, occlusion site - distal M1 aOR 2.5 (1.2–5.1), proximal M1 aOR 5.2 (2.1–13.1), tandem aOR 4.5 (2.1–9.7), ICA-T/L aOR 16.0 (5.7–44.9) in Model 2, adjusted for sex, NIHSS, SBP, on-site EVT facility, occlusion site, thrombus length - Thrombus ≥9 mm: aOR 3.2 (95% CI 1.8–5.7, P=0.002) in Model 2, adjusted for sex, NIHSS, SBP, on-site EVT facility, occlusion site, thrombus length - Of note, perfusion parameters (Tmax volume, HIR) not associated with END.
Shi et al. 2025	- Highest quartile of admission SBP: aOR 2.47 (95% CI 1.47–4.29, P<0.001) - Highest quartile of 24h SBP variability: aOR 2.57 (95% CI 1.34–5.18, P<0.001) - Admission SBP >150 mmHg (ROC cutoff): AUC 0.822 (95% CI 0.749–0.895), sens 73.8%, spec 79.0%, PPV 32.6%, NPV 95.6% - SBP variability >24 mmHg (ROC cutoff): AUC 0.718 (95% CI 0.624–0.811), sens 73.8%, spec 62.9%, PPV 21.5%, NPV 94.6% - Combined SBP + SBP variability: AUC 0.848 (95% CI 0.775–0.922), sens 85.7%, spec 87.2%, PPV 47.2%, NPV 98.8%	Tmax>6s volume highest quartile: aOR 2.09 (95% CI 1.28–5.89, P=0.001); Tmax>6s volumes categorized into quartiles with first quartile as reference.
Wang et al. 2022	NFL - aOR 1.178 (1.038–1.323, P=0.006) in Model 1, adjusted for age and sex - aOR 1.168 (1.034–1.320, P=0.012) in Model 2, adjusted for age, sex, NIHSS, TOAST classification, event to blood sampling (h), and infarct volume (Model 2) - AUC 0.850 (95% CI 0.731–0.970, P<0.001), sensitivity 73.7%, specificity 80% for NFL>55.03 pg/mL (cutoff)	None